

# Mathematical and Computational Methods

## Exercise Sheet 5: Matrices

This sheet accompanies *Unit 5 (Matrices)*. It exercises matrix algebra and multiplication, the determinant and inverse, and the solution of linear systems; it matches the notation of Unit 5 and needs nothing beyond Units 1–5.

**How to use this sheet.** Problems are graded by difficulty—★ drill (routine fluency), ★★ core (standard, tutorial level) and ★★★ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ►) and **further practice** for independent home study.

**Solutions.** A full worked solution sits after each problem: hidden in the student build, shown in the solutions build. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showolutions{}\input{MCM_Exercises_Unit5}"`) and recompile.

### 1 Warm-up drills

Short routine problems, at least one per skill. Keep the arithmetic tidy and check each answer.

#### Matrix algebra

**Exercise 1.1** (★ drill — *addition, scalar multiple and product*). Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 4 \\ -2 & 5 \end{pmatrix}$ . Compute  $\mathbf{A} + \mathbf{B}$ ,  $\mathbf{A} - \mathbf{B}$ ,  $3\mathbf{A}$  and  $2\mathbf{A} - \mathbf{B}$ .

**Exercise 1.2** (★ drill — *which products are defined*). Matrices have sizes  $\mathbf{A}: 2 \times 3$ ,  $\mathbf{B}: 3 \times 3$ ,  $\mathbf{C}: 3 \times 1$ ,  $\mathbf{D}: 1 \times 2$ . For each product  $\mathbf{AB}$ ,  $\mathbf{BA}$ ,  $\mathbf{BC}$ ,  $\mathbf{CD}$ ,  $\mathbf{DA}$ ,  $\mathbf{AC}$ , state whether it is defined and, if so, its size.

**Exercise 1.3** (★ drill — *a  $2 \times 2$  product*). Compute the product  $\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 2 & -2 \end{pmatrix}$ .

**Exercise 1.4** (★ drill — *transpose and trace*). For  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}$  write down  $\mathbf{A}^T$  and compute  $\text{tr}(\mathbf{A})$ .

#### Determinants and inverses

**Exercise 1.5** (★ drill —  *$2 \times 2$  determinants*). Evaluate the three determinants

$$\begin{vmatrix} 3 & 5 \\ 1 & 2 \end{vmatrix}, \quad \begin{vmatrix} 4 & 2 \\ 6 & 3 \end{vmatrix}, \quad \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix}.$$

**Exercise 1.6** (★ drill — *a  $2 \times 2$  inverse*). Find the inverse of  $\mathbf{M} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$  and verify  $\mathbf{MM}^{-1} = \mathbf{I}$ .

**Exercise 1.7** (★ drill — *cofactor expansion*). Evaluate  $\det \begin{pmatrix} 2 & 0 & 1 \\ 3 & 1 & -1 \\ 0 & 2 & 4 \end{pmatrix}$  by cofactor expansion along the first row.

## Linear systems

**Exercise 1.8** (*★ drill — solve by the inverse*). Solve by the inverse method:  $3x + 5y = 1$ ,  $x + 2y = 1$ . (You may reuse  $\mathbf{M}^{-1}$  from Exercise 1.6.)

**Exercise 1.9** (*★ drill — solve by elimination*). Solve by elimination (write the augmented matrix and reduce):  $x + y = 5$ ,  $2x - y = 1$ .

## Transformations

**Exercise 1.10** (*★ drill — transformation matrices*). Write down the  $2 \times 2$  matrices for (a) a rotation by  $90^\circ$  anticlockwise, (b) reflection in the  $x$ -axis, (c) an enlargement by factor 3. Then apply the rotation to  $\mathbf{v} = (2, 1)$ .

## 2 Tutorial problems

The core set for the 2-hour class. These are anchored on the worked examples of Unit 5 and spread applied context across the three cohorts. Standard difficulty unless marked. Problems flagged ► are the live tutorial core to work through in class; the remaining problems in this section round out the topic and are for completion at home.

**Exercise 2.1** (► *★★ core —  $\mathbf{AB} \neq \mathbf{BA}$* ). Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ .

- Compute  $\mathbf{AB}$  and  $\mathbf{BA}$  and confirm they differ.
- Verify  $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$  for these matrices.

**Exercise 2.2** (► *★★ core — data science*). A shop records units sold of three products on two days as the rows of  $\mathbf{S} = \begin{pmatrix} 4 & 2 & 1 \\ 0 & 3 & 5 \end{pmatrix}$  (rows = days, columns = products). The unit prices are  $\mathbf{p} = (3, 5, 2)^T$ .

- Explain why  $\mathbf{Sp}$  is defined and what its entries mean.
- Compute the daily revenue vector  $\mathbf{Sp}$ .

**Exercise 2.3** (*★★ core — determinant and independence*). Let  $\mathbf{T} = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}$ , whose columns are

the vectors  $\mathbf{u} = (1, 2, 0)$ ,  $\mathbf{v} = (0, 1, 1)$ ,  $\mathbf{w} = (1, 0, 2)$ .

- Compute  $\det \mathbf{T}$  by cofactor expansion.
- The volume of the parallelepiped on  $\mathbf{u}$ ,  $\mathbf{v}$ ,  $\mathbf{w}$  is the scalar triple product  $|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$  (Unit 4). Confirm it equals  $|\det \mathbf{T}|$ .

**Exercise 2.4** (► *★★ core — the threaded system, part 1*). Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$ .

- Show  $\det \mathbf{A} = 1$ .
- Find  $\mathbf{A}^{-1}$  by the cofactor (adjugate) method and verify  $\mathbf{AA}^{-1} = \mathbf{I}$ .

We reuse this matrix in Exercise 2.5.

**Exercise 2.5** (► ★★ core — *the threaded system, part 2: two roads*). Solve  $\mathbf{Ax} = \mathbf{b}$  with  $\mathbf{A}$  from Exercise 2.4 and  $\mathbf{b} = (6, 5, 11)^T$ , that is

$$\begin{aligned}x + 2y + 3z &= 6, \\y + 4z &= 5, \\5x + 6y &= 11,\end{aligned}$$

twice: (a) by the inverse method, and (b) by Gaussian elimination on the augmented matrix. Confirm the two answers agree.

**Exercise 2.6** (► ★★ core — *infinitely many solutions*). Solve, by elimination,

$$\begin{aligned}x + 2y - z &= 3, \\2x + y + z &= 3, \\x - y + 2z &= 0,\end{aligned}$$

and write the solution set as the vector equation of a line (Unit 4). How does the echelon form reveal that the system has infinitely many solutions?

**Exercise 2.7** (★★ core — *spot the case*). Apply elimination to

$$x + y + z = 1, \quad x + 2y + 3z = 2, \quad 2x + 3y + 4z = 5,$$

and explain what the echelon form tells you about the solution set.

**Exercise 2.8** (► ★★ core — *physics*). A force  $\mathbf{F} = (2, 3)$  (in newtons) is first rotated  $90^\circ$  anticlockwise and then reflected in the  $x$ -axis.

- Find the single  $2 \times 2$  matrix  $\mathbf{M}$  representing “rotate, then reflect”.
- Compute the transformed force  $\mathbf{MF}$ .
- What single geometric transformation is  $\mathbf{M}$ ?

**Exercise 2.9** (★★ core — *determinant, area and orientation*). For each  $2 \times 2$  transformation matrix below, compute the determinant and state its effect on area and orientation, and whether the map is invertible:

$$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{N} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

**Exercise 2.10** (► ★★ core — *conceptual, true or false*). Decide whether each statement is true or false; give a one-line reason or counterexample.

- Matrix multiplication is commutative.
- Every square matrix has an inverse.
- $(\mathbf{AB})^T = \mathbf{A}^T \mathbf{B}^T$ .
- If  $\det \mathbf{A} = 0$  then  $\mathbf{Ax} = \mathbf{b}$  cannot have a unique solution.
- $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ .

**Exercise 2.11** (★★ core — *maths*). Let  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ . Prove directly that  $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ .

**Exercise 2.12** (★★core — data science: Markov chains (foreshadows Unit 6)). A simple weather model has two states, sunny (S) and rainy (R). The *transition matrix* (columns = today, rows = tomorrow)

$$\mathbf{P} = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix}$$

reads: if today is sunny, tomorrow is sunny with probability 0.8 and rainy with 0.2 (top and bottom of the first column), and similarly the second column handles a rainy today. Today is sunny, so the state vector is  $\mathbf{x}_0 = (1, 0)^\top$ .

- Compute  $\mathbf{x}_1 = \mathbf{P}\mathbf{x}_0$  and say what its entries mean.
- Compute  $\mathbf{P}^2$ , then  $\mathbf{x}_2 = \mathbf{P}^2\mathbf{x}_0$ . Interpret the (1, 1) entry of  $\mathbf{P}^2$  (the probability of being sunny two days after a sunny day).

### 3 Further practice (home)

A larger bank for independent study, running from routine to harder, ending with a few challenge/extension problems marked ★★.

#### Matrix algebra

**Exercise 3.1** (★★core — a  $3 \times 3$  by  $3 \times 2$  product). Compute  $\mathbf{AB}$  where  $\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 0 & 2 & 1 \end{pmatrix}$  and

$$\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

State the size of the result first.

**Exercise 3.2** (★★core — inverse of the transpose). Verify the “inverse of the transpose” rule  $(\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top$  for  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$ . (This is a genuinely different identity from the product–transpose rule  $(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$  that you already meet in Exercises 2.1 and 2.10.)

**Exercise 3.3** (★★core — powers of a shear). Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ . Compute  $\mathbf{A}^2$  and  $\mathbf{A}^3$ , then conjecture a formula for  $\mathbf{A}^n$ .

**Exercise 3.4** (★★core — a symmetric combination). Show that for any square matrix  $\mathbf{A}$  the matrix  $\mathbf{A} + \mathbf{A}^\top$  is symmetric, and illustrate with  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$ .

#### Determinants and inverses

**Exercise 3.5** (★★core — inverse, or none). Find the inverse of each, or state that none exists:

$$(a) \begin{pmatrix} 4 & 7 \\ 1 & 2 \end{pmatrix}, \quad (b) \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}, \quad (c) \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}.$$

**Exercise 3.6** (★★core — choosing the easy row). Evaluate  $\det \begin{pmatrix} 2 & 1 & 3 \\ 0 & 4 & 0 \\ 5 & 6 & 1 \end{pmatrix}$ , choosing a row or column to minimise work.

**Exercise 3.7** (★★ core — *inverse of a triangular matrix*). Find the inverse of  $\mathbf{U} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$  and verify  $\mathbf{U}\mathbf{U}^{-1} = \mathbf{I}$ .

**Exercise 3.8** (★★ core — *when is it singular?*). For which value(s) of  $k$  is each matrix singular?

$$(a) \begin{pmatrix} 1 & k \\ k & 4 \end{pmatrix}, \quad (b) \begin{pmatrix} 1 & 2 & 3 \\ 0 & k & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

## Linear systems

**Exercise 3.9** (★★ core — *solve by the inverse*). Solve by the inverse method, reusing  $\mathbf{U}^{-1}$  from Exercise 3.7:

$$x + y + z = 4, \quad y + z = 2, \quad z = 1.$$

**Exercise 3.10** (★★ core — *Gaussian elimination*). Solve by Gaussian elimination:

$$x + y + z = 6, \quad 2x - y + z = 3, \quad x + 2y - z = 2.$$

**Exercise 3.11** (★★ core — *Gauss–Jordan*). Take the row-echelon form reached in Exercise 3.10,

$$\left( \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & -7 & -21 \end{array} \right),$$

and continue the row operations to *reduced* row-echelon form (leading 1s, with zeros both below and above each pivot). Read the solution off directly, without any back-substitution.

**Exercise 3.12** (★★ core — *conceptual*). Each augmented matrix below is already in echelon form. Classify the system as having a unique solution, no solution, or infinitely many.

$$(a) \left( \begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 4 \end{array} \right), \quad (b) \left( \begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 \end{array} \right), \quad (c) \left( \begin{array}{ccc|c} 1 & 0 & -2 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

**Exercise 3.13** (★★ core — *a one-parameter solution set*). Find all solutions of  $x + y + z = 3$ ,  $x - y + z = 1$  and write the answer as the vector equation of a line.

**Exercise 3.14** (★★ core — *physics*). Mesh analysis of a two-loop circuit gives the currents  $I_1, I_2$  (in amperes) from

$$5I_1 - 2I_2 = 1, \quad -2I_1 + 4I_2 = 6.$$

Solve by the inverse method.

## Transformations

**Exercise 3.15** (★★ core — *reflections*). Write the matrix for reflection in the line  $y = x$ , and use it to find the image of  $(3, 5)$ . Write also the matrices for reflection in the  $y$ -axis and for a  $180^\circ$  rotation.

**Exercise 3.16** (★★ core — *composing transformations*). Let  $\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$  (a horizontal stretch) and  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  (rotation  $90^\circ$ ). Find the matrix for “stretch first, then rotate”, and its determinant. Does the order matter?

**Exercise 3.17** (*★★core — sketch it*). Let  $\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$  (a horizontal stretch) and  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  (rotation by  $90^\circ$ ). On graph paper, sketch the unit square with corners  $(0, 0)$ ,  $(1, 0)$ ,  $(1, 1)$ ,  $(0, 1)$ , then sketch its image under  $\mathbf{D}$  and its image under  $\mathbf{RD}$  (stretch first, then rotate). In each case find the area of the image and check that it equals  $|\det|$  times the original area.

### Challenge / extension

**Exercise 3.18** (*★★★challenge — inverse of a product*). Suppose  $\mathbf{A}$  and  $\mathbf{B}$  are invertible  $n \times n$  matrices. Prove that  $\mathbf{AB}$  is invertible with  $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ .

**Exercise 3.19** (*★★★challenge — links to Unit 3*). Let  $\mathbf{R}(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ . By computing  $\mathbf{R}(\theta)\mathbf{R}(\phi)$ , show that  $\mathbf{R}(\theta)\mathbf{R}(\phi) = \mathbf{R}(\theta + \phi)$ , and read off the angle-addition formulae for  $\cos$  and  $\sin$ .

**Exercise 3.20** (*★★★challenge — commuting matrices*). Find all  $2 \times 2$  matrices  $\mathbf{X}$  that commute with  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , i.e.  $\mathbf{XJ} = \mathbf{JX}$ .

**Exercise 3.21** (*★★★challenge — spot the error*). A student “inverts”  $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$  by writing  $\mathbf{A}^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/4 \end{pmatrix}$  “because you invert each entry,” and then claims the same recipe gives  $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 1/2 \\ 1/3 & 1/4 \end{pmatrix}$ . Identify the error and give the correct inverse of  $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ .

**Exercise 3.22** (*★★★challenge — idempotent and invertible*). Suppose a square matrix satisfies  $\mathbf{A}^2 = \mathbf{A}$  (it is *idempotent*) and is also invertible. Prove that  $\mathbf{A} = \mathbf{I}$ .